

# MANGESH BIDE

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## SUMMARY

Backend / Platform Engineer building production healthcare systems with Python, TypeScript, PostgreSQL, and AWS. Led a greenfield authentication and authorization platform from architecture through production deployment using self-hosted Keycloak, custom Java SPIs, OIDC, multi-tenant SSO with IdP federation, and SpiceDB-based ReBAC. Owns backend architecture and infrastructure end-to-end, with additional experience building real-time voice and agentic AI systems.

## SKILLS

**Languages:** Python, TypeScript, JavaScript (ES6+), Java

**Identity & Authorization:** Keycloak (Organizations, custom SPIs, providers, extensions), SpiceDB, ReBAC, OAuth 2.0, OIDC, JWT, SSO, IdP Federation, Multi-Tenant Architecture, AWS Cognito

**Backend:** Django (DRF, ViewSets), FastAPI, NestJS, Node.js (Express)

**Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB

**Cloud & Infrastructure:** AWS (ECS, EC2, S3, Lambda, API Gateway, IAM, Cognito, CloudWatch), Terraform, Docker, GitHub Actions, CI/CD, Shell Scripting

**Distributed Systems:** Transactional Outbox, Async Workers, Idempotency, Reconciliation, Event-Driven Systems

**Observability:** Sentry, Grafana, ClickHouse, Structured Logging

**AI / Voice:** Gemini, OpenAI, LLM Orchestration, RAG, Embeddings, Agentic AI, Tavus CVI, Daily/WebRTC, Azure Speech

**Frontend:** React.js, Next.js, Tailwind CSS, shadcn/ui

**Practices:** REST API Design, Pytest, Unit/Integration Testing, Code Review, Agile/Scrum

## EXPERIENCE

### Software Development Engineer – I

Feb 2025 – Present

House Works Technology — Remote (promoted from intern, Apr 2025)

- Led a greenfield authentication and authorization platform end-to-end, owning architecture, authorization modeling, service integration, rollout, and production deployment.
- Built and operated self-hosted Keycloak in production, including deployment, configuration, upgrades, backups, and operational maintenance; developed custom Java SPIs, providers, and extensions for business-specific identity requirements.
- Designed fine-grained authorization using SpiceDB (Google Zanzibar model), defining a ReBAC model spanning 16 object types and 45 permissions and integrating authorization checks across Django/DRF APIs.
- Implemented multi-tenancy with Keycloak Organizations and tenant-aware authentication; built IdP Federation with domain-based SSO to route users to their enterprise identity provider based on email domain.
- Kept the SpiceDB authorization graph consistent with PostgreSQL using a transactional outbox and asynchronous worker, with reconciliation and consistency mechanisms to detect and repair authorization drift.
- Architected a real-time multilingual voice transcription and translation pipeline in a production EHR, streaming audio through Azure Speech into structured clinical records in PostgreSQL and serving 1,000+ patients.
- Own the backend stack (Django, DRF, PostgreSQL, AWS) handling 10K–100K requests/day; designed production infrastructure with Terraform, Docker, and AWS EC2 and GitHub Actions CI/CD, reducing deployment time from ~45 min to ~12 min (~70%).
- Set up observability with Sentry, ClickHouse, and Grafana; migrated legacy Django views to DRF ViewSets (~40% less boilerplate), write Pytest unit/integration tests, review code, and ship React.js/Next.js features end-to-end.

### Software Development Engineer Trainee

Oct 2024 – Feb 2025

Procedure Technologies, Mumbai, Maharashtra

- Built full-stack production features using React.js, Next.js, TypeScript, and NestJS; containerized services with Docker and provisioned cloud infrastructure with Terraform.

### Backend Developer Intern

Feb 2024 – Jul 2024

Netwin Infosolutions Pvt Ltd, Nashik, Maharashtra

- Designed REST APIs using Django and MySQL with JWT authentication and RBAC for a Flutter activity-tracking application; integrated ML-powered image search and OTP/email verification.

## PROJECTS

### Mira — Voice Front-Desk Agent | Tavus CVI, FastAPI, Next.js, Supabase, Gemini

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- Built a real-time conversational AI agent with Daily/WebRTC voice interaction, avatar, 7 booking tools, slot-conflict and idempotency guards, 10 unit tests, and Gemini-generated call summaries.

### Fieldnotes — Agentic Job Research System | FastAPI, Gemini, Cloudflare BGE, GitHub API

fieldnotes.mangeshbide.tech

- Built a multi-stage planner-executor-synthesizer agent that researches companies, inspects GitHub engineering footprints, compares resume embeddings against job requirements, and returns structured apply/consider/skip verdicts with cost and latency observability.

## EDUCATION

**B.Tech in Computer Engineering**, R. C. Patel Institute of Technology, Shirpur

2021 – 2024 | CGPA: 7.98/10

**Diploma in Computer Engineering**, R. C. Patel Polytechnic, Shirpur

2019 – 2021 | 93.89%